Feature Engineering for Medicinal Leaf Classification: Ethical Foundations, Visualization-Guided Audits, and Model Evaluation

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***Abstract*—Medicinal plants form a critical part of global healthcare systems. Manual identification of leaf species is labor-intensive and susceptible to human error. This paper presents an interpretable feature engineering pipeline for automatic classification of medicinal leaves using the Indian Medicinal Leaf Image Dataset. We extract color, texture, and shape-based descriptors, apply dimensionality reduction, and evaluate multiple machine learning models. Ethical considerations—including fairness, transparency, and responsible use of data—are embedded throughout the workflow. Experimental results show that Support Vector Machines outperform Logistic Regression and Balanced Random Forests, though challenges such as class imbalance and data variability remain.**

I. Introduction

Medicinal leaves play a vital role in traditional and modern therapeutic practices. Accurate identification is essential for research, pharmacological applications, and biodiversity monitoring. Automated classification through computer vision can reduce misidentification and increase scalability.

Our approach is grounded in ethical AI principles. We emphasize fairness, transparency, and reproducibility while ensuring that modeling decisions remain interpretable. The dataset is used in accordance with open-source policies, and preprocessing steps are carefully documented to allow auditability. The objective is to strike a balance between reliable classification and responsible technology deployment, especially in resource-constrained settings.

II. Background

The Indian Medicinal Leaf Image Dataset contains 5,224 images across 80 classes, each depicting a distinct plant species. The dataset presents several challenges:

- **Severe Class Imbalance:** Some classes contain fewer than 10 samples, while others exceed 140.
- **Lighting Variability:** Uneven illumination leads to inconsistent color profiles.
- **Background Complexity:** Non-uniform backgrounds complicate segmentation and shape extraction.

A. Related Work

Previous studies highlight the effectiveness of handcrafted descriptors in plant identification. Texture features—especially Gray-Level Co-occurrence Matrix (GLCM) and Local Binary Patterns (LBP)—have demonstrated strong performance in leaf-based classification. Shape features are also widely used due to their biological relevance. Hybrid feature engineering approaches, which combine multiple cues, further improve classification accuracy. Interpretability-driven frameworks have gained attention for their ability to explain model decisions [1]–[5].

III. Personal Position

We deliberately adopt a feature engineering strategy instead of a deep learning approach. While convolutional neural networks (CNNs) typically achieve high accuracy, they often lack interpretability. Feature engineering enables explicit control over the type of information extracted from each image, allowing us to understand the role of color, texture, and shape in classification. This approach is also computationally efficient and well-suited for low-power deployment environments. Moreover, interpretability aligns with ethical AI principles by ensuring transparency in decision-making processes.

IV. Data Visualization and Feature Engineering

A. Dataset Exploration

Images were resized to 224×224 pixels and normalized. Data augmentation (horizontal/vertical flips, rotations up to 0.2 radians, and zoom of 0.2) increased diversity and reduced overfitting. Text-based label distribution summaries for training (4,180 images) and validation (1,044 images) confirmed substantial class imbalance.

B. Feature Extraction

We extracted complementary feature categories:

- **Color Features:** Mean RGB values and channel-wise histograms to capture dominant hues and intensity distributions.
- **Texture Features:** GLCM descriptors such as contrast, dissimilarity, homogeneity, energy, correlation,

and angular second moment, computed across multiple orientations.

- **Shape Features:** Contour-based metrics (area, perimeter, aspect ratio, solidity, eccentricity), obtained after Otsu thresholding.

Each image produced a 56-dimensional feature vector.

C. Dimensionality Reduction and Imbalance Handling

Principal Component Analysis (PCA) reduced redundancy and retained 95% of the variance. Because many classes had too few samples to oversample safely, a hybrid resampling strategy was applied. SMOTE was used only for classes with at least two samples, while RandomOverSampler improved representation of remaining underrepresented classes. Text-based audits confirmed improved class balance.

D. Model Evaluation

We evaluated Logistic Regression, Support Vector Machine (linear kernel), and Balanced Random Forest models. Metrics included accuracy, macro-precision, macro-recall, macro-F1, and per-class reports.

TABLE I
VALIDATION PERFORMANCE OF CLASSIFIERS

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	0.3103	0.3166	0.3103	0.2889
SVM (Linear)	0.3554	0.3728	0.3554	0.3403
Balanced Random Forest	0.1216	0.2000	0.1200	0.1100

E. Main Observations

Three key insights emerged:

1) PCA improved model stability by minimizing noise and feature redundancy.
2) SVM demonstrated superior performance, benefiting from a well-compacted feature space.
3) Balanced Random Forest struggled due to high intra-class similarity and limited separability for rare classes.

V. REBUTTAL

A feature engineering pipeline offers interpretability and efficiency but may limit performance on highly diverse image datasets. Deep learning models, while less interpretable, can learn complex spatial hierarchies and may achieve higher accuracy. A promising direction is a hybrid model that fuses lightweight CNN embeddings with handcrafted features. Additionally, class-weighted loss functions or focal loss could mitigate imbalance more effectively. Future research can further refine segmentation techniques to stabilize shape-related descriptors.

VI. CONCLUSION

This study presents an interpretable feature engineering framework for medicinal leaf classification. By combining color, texture, and shape features with PCA and imbalance-aware sampling, we evaluated multiple classical machine learning models. SVM achieved the strongest performance (accuracy 35.54%, F1-score 34.03%). Although class imbalance and inter-class similarity pose significant challenges, the pipeline remains transparent, reproducible, and suitable for low-resource settings. Future work includes exploring hybrid CNN-feature methods and improved imbalance mitigation strategies while maintaining an ethical and transparent design philosophy.

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